Research Proposal

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Preliminary title:
Coercion and Vulnerability: A District-Level Analysis of Rwanda's Crop Intensification Program (CIP) and smallholder resilience to climatic shocks.

Preliminary research question:
Does higher district-level land use consolidation intensity under Rwanda's Crop Intensification Program correlate with greater household food insecurity during climatic shocks?

food insecurity/regional welfare... need to figure out what to use for 'social resilience'.. inhibiting sovereignty over land use, decreasing livelihood flexibility, and constricting resource access. - (clay et al)

[Subquestions]

- Is the share of plots under land use consolidation (LUC) per district negatively associated with household dietary diversity scores when interacted with below-average seasonal rainfall?
- Does the negative interaction between LUC intensity and rainfall shocks increase monotonically as household wealth decreases, indicating that the poorest households bear the greatest resilience cost?
- *Correlation with lower emissions?*

Preliminary knowledge:

Puzzle and relevance. Rwanda's Crop Intensification Program (CIP), launched in 2007, is one of the most coercively enforced agricultural transformation programmes in sub-Saharan Africa. Under the programme's land use consolidation (LUC) component, farmers were mandated to cultivate state-assigned monocrops on consolidated plots, with compliance enforced through Rwanda's *imihigo* performance contract system—a cascade of accountability running from the president through ministers and mayors down to individual farmers (Cioffo et al., 2016). The programme produced nationally reported yield increases for priority crops and is widely cited as a demonstration of what strong state capacity can achieve in agricultural development (Cantore, 2011).

Yet this aggregate productivity narrative conceals a distributional question: *who paid the price for these gains?* Del Prete et al. (2019), using Rwanda's Integrated Household Living Conditions Survey (EICV), find that land consolidation increased production of roots and tubers while reducing dietary diversity—specifically decreasing consumption of meat, fish, fruits, and micronutrient-rich foods. Clay and Zimmerer (2020) go further, arguing through multi-season fieldwork that the CIP reduced smallholder *resilience* by stripping farmers of the crop diversity, livelihood flexibility, and resource access that previously buffered them against climatic variability. Their claim—that CIP households are more exposed to food insecurity when rainfall fails—has not been tested quantitatively. This study addresses that gap.

Theoretical approach. The core argument is that Rwanda's *imihigo* enforcement machinery enabled the state to impose a macro-level productivity goal by removing the micro-level adaptive strategies smallholders use to manage risk. In a democratic context, affected households could resist mandated monocropping through political voice or exit; Rwanda's enforcement cascade foreclosed both options. The result is a structural vulnerability: CIP households are locked into state-assigned crops with reduced ability to shift cultivation in response to adverse weather—a cost that falls hardest on those with the fewest alternative resources.

Hypotheses.

- **H1** Households in districts with higher LUC intensity will show lower dietary diversity scores during years of below-average rainfall than households in low-intensity districts (*resilience-reducing effect of CIP*).
- **H2** The vulnerability-amplifying effect of LUC intensity will increase as household wealth decreases (*poverty-concentration hypothesis*).

Operationalization. The key independent variable is **district-level LUC intensity**, defined as the share of sampled agricultural plots recorded as located within an official land consolidation site in a given season. This is constructed from variable `s2q12` ("Is this plot located in a land consolidation site?") in the NISR Seasonal Agricultural Survey (SAS) plot-level microdata, aggregated to the district level. The primary dependent variable is **household dietary diversity score (DDS)**, constructed from the food consumption module of the EICV household survey. The **climatic shock variable** is the standardised seasonal rainfall anomaly per district, derived from CHIRPS v2.0 gridded precipitation data aggregated to district polygons. The key estimating quantity is the interaction between LUC intensity and rainfall anomaly. The unit of analysis is the **household**, nested within districts ($N \approx 14{,}000$ households across 30 districts).

Preliminary structure:

1. Introduction
2. Overview of the relevant literature (focus on Clay & Zimmerer, 2020; Del Prete et al., 2019)
3. Theory and hypotheses: The Trade-off between Specialisation and Resilience
4. Empirics
   - 4.1 Research design and operationalization
   - 4.2 Methodology
   - 4.3 Results
5. Conclusion

Methodology note:
The study employs OLS regression with district fixed effects, estimated on a cross-sectional household sample from the EICV4 (2013/14) or EICV5 (2016/17) survey—the waves covering the period of mature CIP implementation. The district-level LUC intensity variable is merged to the household data via the common district identifier. The baseline model is:

$$\mathrm{DDS}_{ihd} = \beta_1 \,\mathrm{LUC}_d + \beta_2 \,\mathrm{Rain}_d + \beta_3 \big(\mathrm{LUC}_d \times \mathrm{Rain}_d\big) + \mathbf{Z}'_{ih}\boldsymbol{\gamma} + \alpha_d + \varepsilon_{ihd} \tag{1}$$

where DDS_{ihd} is the dietary diversity score for household h in district d; LUC_d is the district share of plots under land consolidation; Rain_d is the standardised rainfall anomaly; β_3 is the key coefficient (a negative value supports H1); $\mathbf{Z}_{ih}$ is a vector of household-level controls (land area, market distance, household size, asset index); α_d are district fixed effects; and ε_{ihd} is the error term clustered at the district level. To test H2 and H3, the model is extended with a triple interaction term $\text{LUC}_d \times \text{Rain}_d \times M_{ih}$, where M_{ih} is female household headship or wealth quintile. Propensity score matching (PSM) is used as a robustness check.

Data sources:

Variable	Source	Access
Dietary diversity score (DV)	EICV4 (2013/14) or EICV5 (2016/17)	World Bank Microdata Catalog; free registration
LUC intensity (% plots; `s2q12`)	NISR Seasonal Agricultural Survey microdata	IHSN catalog / NISR microdata catalog; free registration
Seasonal rainfall anomaly	CHIRPS v2.0 gridded precipitation	data.chc.ucsb.edu; direct download, no registration
District merge key	Common to EICV and SAS	Included in both datasets

Data construction note. The LUC intensity variable is constructed in two steps: (1) download the NISR SAS plot-level microdata and extract variable `s2q12`; (2) collapse to district level to obtain the share of plots under LUC per district per season. This measure is merged to the EICV household data via the district code. CHIRPS raster files are aggregated to Rwanda's 30 district polygons and the seasonal anomaly is computed as the standardised deviation from the 1981–2010 district mean.

Literature:

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